2024, Mar. 2024 Release LOC:Acute Adult
Gastrointestinal (GI) Bleeding

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview
Select Day
- **Initial review** (1, 2, 3, 4)
- **Episode Day 1** (1, 3, 4)
- **Episode Day 2** (3, 4, 35)
- **Episode Day 3** (3, 4)
- **Episode Day 4**

Discharge Screens (49)
Utilization Benchmarks
Notes

Overview

Instruction:
This subset is for patients with acute gastrointestinal (GI) bleeding (upper and lower), including GI bleeding post-ambulatory procedure (e.g., post-polypectomy bleed).
Introduction:
Gastrointestinal (GI) bleeding is divided into upper or lower, defined by its location relative to the ligament of Treitz in the terminal duodenum. Bleeding originating in the esophagus, stomach, and duodenum is considered upper, and all others are lower. While upper GI bleeding is more common than lower, both are common medical emergencies. Typical causes of upper GI bleeding include peptic ulcers, gastric erosion, esophageal varices, Mallory-Weiss tears, angiodysplasia, esophagitis, and gastric cancer. Typical causes of lower GI bleeding include diverticular disease, colitis, angiodysplasia, cancer, and upper GI bleeding. Upper GI bleeding commonly produces bloody or coffee ground like vomit (hematemesis), or black, tarry stools (melena). Lower GI bleeding commonly produces bright red or maroon-colored blood per rectum (hematochezia). Symptoms may include abdominal pain, nausea, vomiting, tachycardia, dyspnea, lethargy, or in cases of extreme blood loss, chest pain, confusion, syncope, and other signs of hypovolemic shock. Two causes of GI bleeding which can rapidly progress without immediate intervention are esophageal varices and aorto-enteric fistula (DeGeorge and Nable, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 10 edn. 2023:240-33).
Evaluation and Treatment:
Prompt effort should be made to determine the source of bleeding. A history and physical should include inquiries regarding recent surgical procedures, recent medications (e.g., NSAIDs, aspirin), vital signs, including orthostatics, signs of pallor, jaundice, bowel sounds, palpation of abdomen for masses, and rectal exam. Testing should include laboratory studies (e.g., CBC, BUN, creatinine, coagulation, electrolytes), guaiac stool testing for occult blood, and cross-matching, should blood transfusion become urgently necessary. An endoscopy or colonoscopy may be useful to identify the source of bleeding and allows for certain potential treatment, however, abdominal computed tomography (CT) is the imaging test of choice for patients in the emergency department. An electrocardiogram (ECG) may be warranted in patients over 35 years of age and in those with cardiac risk factors (e.g., diabetes, prior myocardial infarction, tobacco use). CT angiography (CTA) can be useful in finding the exact location of bleeding if endoscopy is unsuccessful. Common medical treatments include fluid resuscitation, proton pump inhibitors, histamine H2-receptor antagonists, antibiotics, octreotide, and anticoagulant reversal agents when appropriate. Various invasive procedures used to stop GI bleeding may include esophageal balloon tamponade, endoscopic banding, angiographic embolization, and transjugular intrahepatic portosystemic shunting (TIPS) (DeGeorge and Nable, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 10 edn. 2023:240-33; Sengupta et al., Am J Gastroenterol 2023, 118: 208-31; Abraham et al., Am J

Gastroenterol 2022, 117: 542-58; Laine et al., Am J Gastroenterol 2021, 116: 899-917; American College of Radiology, ACR Appropriateness Criteria: Radiologic Management of Lower Gastrointestinal Tract Bleeding. 2020).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:

- **Initial review, One:** (1, 2, 3, 4)
  - (From arrival in ED through decision to admit)
  - (Symptom or finding within 24h)
  - **OBSERVATION, One:** (5)
    - Gastrointestinal (GI) bleeding, **Both:**
      - Coffee ground emesis, hematemesis, hematochezia, or melena (6, 7, 8, 9, 10)
      - Hct ≥ 21%(0.21) or Hb ≥ 7.0 g/dL(70 g/L) (11)
  - **ACUTE, ≥ One:** (12)
    - Lower gastrointestinal (GI) bleeding and, **One:** (13)
      - Hematochezia or melena and, ≥ **One:** (9, 10)
        - Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) (11)
        - Exertional dyspnea worse than baseline (14, 15)
        - INR ≥ 2.0
        - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 (16, 17)
        - Orthostatic hypotension, **Both:** (18)
          - Sustained drop in blood pressure within 3 min of sitting or standing
          - Systolic blood pressure drop ≥ 20 mmHg or diastolic blood pressure drop ≥ 10 mmHg
        - Syncope (19)
    - Upper gastrointestinal (GI) non-variceal bleeding and, **One:** (20)
      - Coffee ground emesis, hematemesis, hematochezia, or melena and, ≥ **One:** (6, 7, 8, 9, 10)
        - Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) (11)
        - Exertional dyspnea worse than baseline (14, 15)
        - INR ≥ 2.0
        - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 (16, 17)
        - Orthostatic hypotension, **Both:** (18)
          - Sustained drop in blood pressure within 3 min of sitting or standing
          - Systolic blood pressure drop ≥ 20 mmHg or diastolic blood pressure drop ≥ 10 mmHg
        - Syncope (19)
  - **CRITICAL, ≥ One:** (21)
    - Embolization performed within 24h (22)
    - Variceal bleeding (23)
    - Hemodynamic instability, **All:** (24)
      - Initial treatment, ≥ **One:**
        - ≥ 2 units blood product transfusion
        - ≥ 2 liters IV fluid (25)
        - ≥ 1 unit blood product transfusion and ≥ 1 liter IV fluid
      - Findings after initial treatment, ≥ **One:**
        - Heart rate > 120/min
        - Systolic blood pressure < 100 mmHg and < baseline
        - MAP < 65 mmHg
      - Requiring ongoing resuscitation
    - Mechanical ventilation
    - NIPPV (26)
    - Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 1, One:** (1, 3, 4)
  - (Symptom or finding within 24h)
  - **OBSERVATION, Both:** (5)
    - Gastrointestinal (GI) bleeding, **Both:**
      - Coffee ground emesis, hematemesis, hematochezia, or melena (6, 7, 8, 9, 10)
      - Hct ≥ 21%(0.21) or Hb ≥ 7.0 g/dL(70 g/L) (11)
    - Intervention, ≥ **One:**

- Blood product transfusion
- Hct or Hb monitoring at least 2x/24h

- **ACUTE,** ≥ **One:** (12)
  - Lower gastrointestinal (GI) bleeding and, **Both:** (13)
    - Hematochezia or melena and, ≥ **One:** (9, 10)
      - Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) (11)
      - Exertional dyspnea worse than baseline (14, 15)
      - INR ≥ 2.0
      - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 (16, 17)
      - Orthostatic hypotension, **Both:** (18)
        - Sustained drop in blood pressure within 3 min of sitting or standing
        - Systolic blood pressure drop ≥ 20 mmHg or diastolic blood pressure drop ≥ 10 mmHg
      - Syncope (19)
    - Intervention, ≥ **One:**
      - Blood product transfusion
      - Colonoscopy performed within 24h (27)
      - Hct or Hb monitoring at least 2x/24h
  - Upper gastrointestinal (GI) non-variceal bleeding and, **Both:** (20)
    - Coffee ground emesis, hematemesis, hematochezia, or melena and, ≥ **One:** (6, 7, 8, 9, 10)
      - Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) (11)
      - Exertional dyspnea worse than baseline (14, 15)
      - INR ≥ 2.0
      - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 (16, 17)
      - Orthostatic hypotension, **Both:** (18)
        - Sustained drop in blood pressure within 3 min of sitting or standing
        - Systolic blood pressure drop ≥ 20 mmHg or diastolic blood pressure drop ≥ 10 mmHg
      - Syncope (19)
    - Hct or Hb monitoring at least 2x/24h and, ≥ **One:**
      - Blood product transfusion
      - Endoscopy performed within 24h (28)
      - High-dose proton pump inhibitor ≤ 3d (29)
- **CRITICAL,** ≥ **One:** (21)
  - Embolization performed within 24h (22)
  - Variceal bleeding and, ≥ **One:** (23)
    - Balloon tamponade (30)
    - Therapeutic endoscopy and, ≥ **One:** (31, 32)
      - Octreotide
      - Somatostatin
    - Transjugular intrahepatic portosystemic shunt (TIPS) (33)
  - Hemodynamic instability, **All:** (24)
    - Initial treatment, ≥ **One:**
      - ≥ 2 units blood product transfusion
      - ≥ 2 liters IV fluid (25)
      - ≥ 1 unit blood product transfusion and ≥ 1 liter IV fluid
    - Findings after initial treatment, ≥ **One:**
      - Heart rate > 120/min
      - Systolic blood pressure < 100 mmHg and < baseline
      - MAP < 65 mmHg
    - Requiring ongoing resuscitation
  - Hemodynamic monitoring, invasive (34)
  - Mechanical ventilation
  - NIPPV (26)
  - Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

- **Episode Day 2, One:** [3, 4, 35]
  - **OBSERVATION, One:**
    - **Responder**, discharge expected today if clinically stable last 12h, **All:** [36]
      - Gastrointestinal (GI) bleeding controlled or resolved
      - Hct or Hb within acceptable limits [37]
      - No complication or active comorbidity relevant to this episode of care [38]
    - **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
      - For gastrointestinal (GI) bleeding use Episode Day 2 at a higher level of care
      - For a condition other than gastrointestinal (GI) bleeding use appropriate subset (Episode Day 1) based on clinical findings
      - Refer for secondary review
  - **ACUTE, One:**
    - **Responder**, discharge expected today if clinically stable last 12h, **All:** [36]
      - Gastrointestinal (GI) bleeding controlled or resolved
      - Hct or Hb within acceptable limits [37]
      - Vital signs within acceptable limits [37]
      - No complication or active comorbidity relevant to this episode of care [38]
    - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
      - Lower gastrointestinal (GI) bleeding and, **Both:**
        - Finding, ≥ **One:**
          - Angiodysplasia and risk of rebleed [39]
          - Hb decrease ≥ 2.0 g/dL(20 g/L) within last 24h [40]
          - Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) [11]
          - Exertional dyspnea worse than baseline [14, 15]
          - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 [16, 17]
          - Orthostatic hypotension, **Both:** [18]
            - Sustained drop in blood pressure within 3 min of sitting or standing
            - Systolic blood pressure drop ≥ 20 mmHg or diastolic blood pressure drop ≥ 10 mmHg
          - Persistent bleeding
          - Syncope [19]
        - Hct or Hb monitoring at least 2x/24h and, ≥ **One:**
          - Blood product transfusion
          - Colonoscopy performed within 24h [27]
      - Upper gastrointestinal (GI) non-variceal bleeding and, **Both:**
        - Finding, ≥ **One:**
          - Hb decrease ≥ 2.0 g/dL(20 g/L) within last 24h [40]
          - Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) [11]
          - Exertional dyspnea worse than baseline [14, 15]
          - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 [16, 17]
          - Orthostatic hypotension, **Both:** [18]
            - Sustained drop in blood pressure within 3 min of sitting or standing
            - Systolic blood pressure drop ≥ 20 mmHg or diastolic blood pressure drop ≥ 10 mmHg
          - Persistent bleeding
          - Risk of rebleed, ≥ **One:** [41]
            - Adherent clot
            - Bleeding disorder
            - Ulcer size ≥ 2 cm
            - Visible non-bleeding vessel
          - Syncope [19]
        - Hct or Hb monitoring at least 2x/24h and, ≥ **One:**
          - Blood product transfusion
          - Endoscopy performed within 24h [42]

- High-dose proton pump inhibitor ≤ 3d [29]
- Post Critical care ≤ 24h

⊖ **INTERMEDIATE,** ≥ **One:** [43]
- NIPPV [26]
- Oxygen ≥ 40%(0.40), ≤ 2d [44]

⊖ **CRITICAL,** ≥ **One:**
- Embolization performed within 24h [22]
- Post embolization ≤ 24h
- ⊖ Variceal bleeding controlled ≤ 2d and, **Both:** [23]
  - Hemodynamic stability [45]
  - Octreotide or somatostatin [32]
- ⊖ Hemodynamic instability persisting after ≥ 1L of IVF resuscitation or ≥ 1 unit blood product transfusion and, **Both:** [24]
  - ⊖ Finding, ≥ **One:**
    - Heart rate > 120/min
    - Systolic blood pressure < 100 mmHg and < baseline
    - MAP < 65 mmHg
  - ⊖ Intervention, ≥ **One:**
    - Blood product transfusion
    - IV fluid resuscitation ≥ 1L ≤ 24h [25]
- Hemodynamic monitoring, invasive [34]
- Mechanical ventilation
- ⊖ NIPPV and, ≥ **One:** [26]
  - FiO_2 > 50%(0.50)
  - Respiratory intervention every 1-2h [46]
- Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

⊖ **Episode Day 3, One:** [3, 4]

⊖ **ACUTE, One:**
- ⊖ **Responder**, discharge expected today if clinically stable last 12h, **All:** [36]
  - Gastrointestinal (GI) bleeding controlled or resolved
  - Hct or Hb within acceptable limits [37]
  - Vital signs within acceptable limits [37]
  - No complication or active comorbidity relevant to this episode of care [38]
- ⊖ **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
  - ⊖ Lower gastrointestinal (GI) bleeding and, **Both:**
    - ⊖ Finding, ≥ **One:**
      - Angiodysplasia and risk of rebleed [39]
      - Hb decrease ≥ 2.0 g/dL(20 g/L) within last 24h [40]
      - Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) [11]
      - Exertional dyspnea worse than baseline [14, 15]
      - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 [16, 17]
      - ⊖ Orthostatic hypotension, **Both:** [18]
        - Sustained drop in blood pressure within 3 min of sitting or standing
        - Systolic blood pressure drop ≥ 20 mmHg or diastolic blood pressure drop ≥ 10 mmHg
      - Syncope [19]
    - ⊖ Intervention, **Both:**
      - Blood product transfusion
      - Hct or Hb monitoring at least 2x/24h
  - ⊖ Upper gastrointestinal (GI) non-variceal bleeding and, **Both:**
    - ⊖ Finding, ≥ **One:**
      - Hb decrease ≥ 2.0 g/dL(20 g/L) within last 24h [40]
      - Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) [11]
      - Exertional dyspnea worse than baseline [14, 15]

- Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 [16, 17]
- Orthostatic hypotension, **Both:** [18]
  - Sustained drop in blood pressure within 3 min of sitting or standing
  - Systolic blood pressure drop ≥ 20 mmHg or diastolic blood pressure drop ≥ 10 mmHg
- Risk of rebleed, ≥ **One:** [41]
  - Adherent clot
  - Bleeding disorder
  - Ulcer size ≥ 2 cm
  - Visible non-bleeding vessel
- Syncope [19]

- Hct or Hb monitoring at least 2x/24h and, ≥ **One:**
  - Blood product transfusion
  - High-dose proton pump inhibitor ≤ 3d [29]

- Upper gastrointestinal (GI) variceal bleeding and, **Both:**
  - Bleeding controlled > 2d and, ≥ **One:**
    - Medium or large varices [47]
    - Red wale marks found on endoscopy [47]
    - Child-Pugh score 9 or higher [47, 48]
  - Hct or Hb monitoring at least 2x/24h and, ≥ **One:**
    - Blood product transfusion
    - Octreotide or somatostatin [32]
- Post Critical care ≤ 24h

- **INTERMEDIATE,** ≥ **One:**
  - NIPPV [26]
  - Oxygen ≥ 40%(0.40), ≤ 2d [44]
- **CRITICAL,** ≥ **One:**
  - Embolization performed within 24h [22]
  - Post embolization ≤ 24h
  - Variceal bleeding controlled ≤ 2d and, **Both:** [23]
    - Hemodynamic stability [45]
    - Octreotide or somatostatin [32]
  - Hemodynamic instability persisting after ≥ 1L of IVF resuscitation or ≥ 1 unit blood product transfusion and, **Both:** [24]
    - Finding, ≥ **One:**
      - Heart rate > 120/min
      - Systolic blood pressure < 100 mmHg and < baseline
      - MAP < 65 mmHg
    - Intervention, ≥ **One:**
      - Blood product transfusion
      - IV fluid resuscitation ≥ 1L ≤ 24h [25]
  - Hemodynamic monitoring, invasive [34]
  - Mechanical ventilation
  - NIPPV and, ≥ **One:** [26]
    - FiO_2 > 50%(0.50)
    - Respiratory intervention every 1-2h [46]
  - Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

- **Episode Day 4, One:**
  - **ACUTE, One:**
    - **Responder**, discharge expected today if clinically stable last 12h, **All:** [36]
      - Gastrointestinal (GI) bleeding controlled or resolved
      - Hct or Hb within acceptable limits [37]
      - Vital signs within acceptable limits [37]
      - No complication or active comorbidity relevant to this episode of care [38]

- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
  - **Use Extended Stay subset**
  - Refer for secondary review
- **INTERMEDIATE, One:**
  - **Non-responder,** not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
    - **Use Extended Stay subset**
    - Refer for secondary review
- **CRITICAL, One:**
  - **Non-responder,** not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
    - **Use Extended Stay subset**
    - Refer for secondary review

Discharge, Level of Care, One: (49)

- **HOME, Both:**
  - Level of care appropriateness, **Both:**
    - Home environment safe and accessible (50)
    - Patient or caregiver demonstrates ability to manage care
  - Gastrointestinal bleeding discharge planning, **All:**
    - Provide comprehensive written discharge and teaching instructions (51)
    - Education on the importance of, **All:**
      - Anticoagulation management
      - Avoidance of foods that may cause gastrointestinal irritation
      - Caution when using nonsteroidal anti-inflammatory medications or aspirin
      - Elimination or reduction of alcohol intake if indicated
      - Follow-up appointments
      - Knowing when to seek medical care
      - Smoking cessation program if indicated (52)
    - Medication reconciliation (53)
    - Follow-up care planned (54)
    - Identify and address transportation needs
- **HOME CARE, Both:**
  - Level of care appropriateness, **All:**
    - Home environment safe and accessible (50)
    - Patient or caregiver willing and able to learn care
    - Outpatient management contraindicated or unavailable (55, 56)
    - Skilled services, ≥ **One:** (57)
      - Skilled nursing (58)
      - Skilled therapy: PT, OT, or ST
      - Behavioral health (59)
  - Gastrointestinal bleeding discharge planning, **All:**
    - Provide comprehensive written discharge and teaching instructions (51)
    - Education on the importance of, **All:**
      - Anticoagulation management
      - Avoidance of foods that may cause gastrointestinal irritation
      - Caution when using nonsteroidal anti-inflammatory medications or aspirin
      - Elimination or reduction of alcohol intake if indicated
      - Follow-up appointments
      - Knowing when to seek medical care
      - Smoking cessation program if indicated (52)
    - Medication reconciliation (53)

- Follow-up care planned [54]
- Identify and address transportation needs

- **BEHAVIORAL HEALTH, Both:**
  - Level of care appropriateness, **One:**
    - Support systems available [60]
  - Skilled treatment, ≥ **One:**
    - Clinical assessment [61]
    - Individual, group, or family counseling
    - Psychiatric, substance, or medication evaluation [62]
- **SKILLED MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 1-2x/24h [58]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [63, 64]
        - Functional impairments requiring at least supervision [65]
        - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [53]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 4h/24h [58]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [63, 64]
        - ≥ 2 functional impairments requiring at least minimum assistance [65]
        - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [53]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **SUBACUTE REHAB, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Treatment precluded in a lower level
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [63, 64]
      - ≥ 2 functional impairments requiring at least minimum assistance [65]
      - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [53]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **ACUTE REHAB, Both:**

- Level of care appropriateness, **Both:**
  - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
  - Services, **All:**
    - Goal directed therapy with rehabilitation potential [63, 64]
    - ≥ 2 functional impairments requiring at least minimum assistance [65]
    - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines [66]
- Complete prior to facility transfer, **All:**
  - Medication reconciliation [53]
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation

LONG TERM ACUTE CARE, Both:

- Level of care appropriateness, **Both:**
  - Treatment precluded in a lower level
  - In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
    - Primary condition or illness and treatment of active comorbid condition(s) [67]
    - Ventilator dependent ≥ 6h/d and weaning planned [68, 69]
    - Acute and comorbid conditions requiring prolonged hospitalization [70]
- Complete prior to facility transfer, **All:**
  - Medication reconciliation [53]
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
368 MAJOR ESOPHAGEAL DISORDERS WITH MCC	**n/a**	**4.2**	**CMS GMLOS**
369 MAJOR ESOPHAGEAL DISORDERS WITH CC	**n/a**	**3**	**CMS GMLOS**
370 MAJOR ESOPHAGEAL DISORDERS WITHOUT CC/MCC	**n/a**	**2.2**	**CMS GMLOS**
377 GASTROINTESTINAL HEMORRHAGE WITH MCC	**n/a**	**4.6**	**CMS GMLOS**
378 GASTROINTESTINAL HEMORRHAGE WITH CC	**n/a**	**3**	**CMS GMLOS**
379 GASTROINTESTINAL HEMORRHAGE WITHOUT CC/MCC	**n/a**	**2.1**	**CMS GMLOS**
380 COMPLICATED PEPTIC ULCER WITH MCC	**n/a**	**5.1**	**CMS GMLOS**
381 COMPLICATED PEPTIC ULCER WITH CC	**n/a**	**3.2**	**CMS GMLOS**
382 COMPLICATED PEPTIC ULCER WITHOUT CC/MCC	**n/a**	**2.4**	**CMS GMLOS**
383 UNCOMPLICATED PEPTIC ULCER WITH MCC	**n/a**	**3.8**	**CMS GMLOS**
384 UNCOMPLICATED PEPTIC ULCER WITHOUT MCC	**n/a**	**2.6**	**CMS GMLOS**
385 INFLAMMATORY BOWEL DISEASE WITH MCC	**n/a**	**5**	**CMS GMLOS**

Condition or Procedure	% Pd Obs	LOS (days)	Type
386 INFLAMMATORY BOWEL DISEASE WITH CC	n/a	3.4	CMS GMLOS
387 INFLAMMATORY BOWEL DISEASE WITHOUT CC/MCC	n/a	2.6	CMS GMLOS
393 OTHER DIGESTIVE SYSTEM DIAGNOSES WITH MCC	n/a	4.4	CMS GMLOS
394 OTHER DIGESTIVE SYSTEM DIAGNOSES WITH CC	n/a	3	CMS GMLOS
395 OTHER DIGESTIVE SYSTEM DIAGNOSES WITHOUT CC/MCC	n/a	2.1	CMS GMLOS
LOWER GI BLEEDING	35%	3.7	InterQual
RUPTURED VARICES	6%	3.8	InterQual
UPPER GI (NON-VARICEAL) BLEEDING	13%	3.9	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:

1:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.

- **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.
For further clarification, see the Acute Level of Care Review Process.

3:

Expected progress after Day 2:

- Bleeding resolved
- Hb and Hct improving or no Hb decrease ≥ 2.0 g/dL(20g/L) within last 24h
- Vital signs stable or within acceptable limits

May also consider:

- Proton pump inhibitors or histamine H2-receptor antagonist
- Discharge planning

If no, Consider:

- Surgical consultation

4:

Potential risk of readmission

- Knowledge deficit or nonadherence with condition management identified (**Home Care, Readmission reduction team**)
- Comorbid cirrhosis, chronic obstructive pulmonary disease (COPD), hypertension, diabetes (**Home Care, Readmission reduction team**)

Functional impairment

- Physical therapy provided in the home setting should be considered if the home is safe and accessible (**Home Care**)
- If home environment is unsafe, consider a skilled nursing facility (**SNF**)

Social Determinants of Health (SDOH)

Social concerns may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. The following social determinants of health should be evaluated on admission to determine which factors are relevant to an individual patient and condition. This list may not be all inclusive, and all relevant SDOH factors should be considered. Any SDOH factors determined to be relevant to a given patient should be planned for throughout the hospital stay.

Alcohol or substance use

- Care coordination with outpatient medical or behavioral health providers
- Information should be given about community resources and programs
- Referral to addiction social support systems

Domestic violence and sexually abused individuals

- Community resources and legal information should be given
- Create a long-term safety plan or make appropriate referral

End of life planning, arrange palliative care or hospice

- Consider palliative care for patients with end-stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

Financial resources strained or limited

- For patients with income insecurity, identify income-support resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Consider direct income assistance benefits and indirect income assistance programs (i.e., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)
- Consider referral to Area Agencies on Aging (AAA) for information on nutrition and meals programs, homemaking services, transportation resources
- For food insecurity, provide information about local food banks and Supplemental Nutrition Assistance Program (SNAP) incentive programs (Berman et al., Pediatr Rev 2018, 39: 235-46)

Functional or activity of daily living (ADL) impairment

- Consider an alternate level of care
- If patient is unable to make decision for self, refuses treatment, or demands to be discharged without a rational reason, a capacity evaluation may be needed to ensure patient's ability to make decisions for self
- Information should be given to patient and family about Advance Directive or Power of Attorney

Mental health comorbidities involved

- Appointment with primary care provider or behavioral health specialist upon discharge
- Care coordination with outpatient behavioral health providers
- Community resources should be given

Risk for readmission

- Consider home care services
- Consider transfer to an alternative level of care or readmission reduction program
- Provide patient and family with information on disease management programs
- Consider telehealth and mobile health interventions

Risk for treatment nonadherence

- For patients with poor insight into illness or noncompliance with treatment, introduce peer support groups to help improve service user experience and quality of life (National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
- In-home intervention to prevent medical relapse, or readmission, and to enhance medication compliance
- For patients at risk of medication nonadherence, consider patient education and behavioral support initiatives such as text message reminders to take a medication and electronic medication dispensers

Environmental exposure to toxic substances and other physical hazards

- For tobacco use and/or exposure, information should be given about tobacco-use cessation and the risk of second-hand smoke
- For lead poisoning, provide community resources for primary prevention and secondary prevention strategies (i.e., blood lead level testing and follow up) (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)

Unstable living environment or homelessness expected upon discharge

- Health insurance application and information should be given to patient
- Homeless shelter application and information should be given to patient
- Referral to walk-in state or federally sponsored outpatient programs
- Ensure access to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
- For individuals with homelessness and the greatest service need (i.e., serious mental illness, history of hospitalizations, functional impairment), consider Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community-Based Treatment (Ponka et al., PLoS One 2020, 15: e0230896)

Unsafe home environment

- If home environment is unsafe, consider a skilled nursing facility (SNF)
- For patients at risk for falls, provide information on community-based fall prevention programs through the National Council on Aging

5:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

6:

Vomiting of dark, coffee ground material is indicative of bleeding from the esophagus, stomach, or duodenum (DeGeorge and Nable, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 10 edn. 2023:240-33).

7:

Hematemesis is defined as vomiting blood, which includes bright red blood, suggesting recent or continued bleeding, or dark, coffee ground material, suggesting a slow bleed or bleeding that has stopped. Hematemesis indicates bleeding from the esophagus, stomach, or duodenum (DeGeorge and Nable, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 10 edn. 2023:240-33).

8:

Instruction: This criterion may be applied for suspected hematemesis (e.g., when a patient reports vomiting blood).

9:

Hematochezia is defined as maroon or bright red blood passed from the rectum and usually signifies a lower gastrointestinal source. In these criteria, hematochezia does not include hemorrhoidal bleeding (DeGeorge and Nable, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 10 edn. 2023:240-33).

10:

Melena is the passage of black or tarry stools from the rectum (DeGeorge and Nable, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 10 edn. 2023:240-33).

11:

Blood transfusion practice guidelines are intended to improve clinical outcomes while avoiding unnecessary transfusions that may not be clinically indicated. Recent high-quality evidence has resulted in guideline updates recommending lower hemoglobin transfusion thresholds for most hospitalized adult and pediatric patients. While individual institutional practice patterns may vary, the American Association of Blood Banks (AABB) recommendation for hemodynamically stable patients, including critically ill patients, is a restrictive transfusion threshold for a hemoglobin of less than 7 g/dL for most patients. A hemoglobin threshold of 7.5 g/dL is recommended for transfusion if patients are undergoing cardiac surgery and a threshold of 8 g/dL if patients are undergoing orthopedic surgery or have pre-existing cardiac disease. There are exceptions to these parameters including patients with acute coronary syndrome, severe thrombocytopenia, and chronic transfusion-dependent anemia (Carson et al., Jama 2023: E1-E11; Chegondi et al., Transfus Med Hemother 2016, 43: 297-301).

12:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

13:

Lower gastrointestinal bleeding requiring treatment at the Acute level of care is defined as acute overt bleeding, causing signs and symptoms of anemia which require further diagnostic testing (Aoki et al., World J Gastroenterol 2019, 25: 69-84).

14:

Patients with severe anemia may exhibit exertional dyspnea (e.g., shortness of breath while walking up a flight of stairs) due to lack of circulating oxygen-carrying red blood cells to muscles. This may require hospitalization for blood transfusion depending on thresholds and possible need for supplemental oxygen (Yui and Brodsky, Hematol Oncol Clin North Am 2022, 36: 325-39; Long and Koyfman, Emerg Med Clin North Am 2018, 36: 609-30; DeVos and Jacobson, Emerg Med Clin North Am 2016, 34: 129-49).

15:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

16:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).
Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

17:

Glasgow Coma Scale (GCS) is a tool commonly used to measure neurological function. GCS evaluates the level of consciousness in acutely ill patients. A patient's GCS is the combined total score accumulated through the evaluation of the following 3 areas (ranging from 3-15) (Chou et al., In: Glasgow Coma Scale for Field Triage of Trauma: A Systematic Review. 2017):

- Eye opening - 4 = Spontaneous, 3 = To voice, 2 = To pain, 1 = None
- Verbal response - 5 = Oriented, 4 = Confused, 3 = Inappropriate, 2 = Incomprehensible sounds, 1 = None
- Motor response - 6 = Obeys verbal commands, 5 = Localizes to pain, 4 = Withdraws to pain, 3 = Flexion response to pain, 2 = Extension response to pain, 1 = None

Of note, age may impact the relationship between anatomic injury severity and GCS scores such that the same GCS score may correspond to a greater severity of traumatic brain injury (TBI) in adults aged 65 and older than in those younger than 65 (Salottolo et al., JAMA Surg 2014, 149: 727-34). For patients less than 5 years of age, the verbal response component includes a modified scoring as follows:

- 5 = Appropriate word or social smile / Coos and babbles, 4 = Cries but consolable, 3 = Cries, screams, persistently irritable, 2 = Restless, agitated, moans, 1 = None

18:

Orthostatic hypotension (Juraschek et al., JAMA Intern Med 2017, 177: 1316-23).

19:

Syncope is the transient loss of consciousness caused by diminished cerebral blood flow, identified as brief, with spontaneous onset and recovery (Brignole et al., Eur Heart J 2018, 39: 1883-948; Shen et al., Journal of the American College of Cardiology 2017, 70: e39-e110).

20:

Non-variceal upper gastrointestinal (GI) bleeds may be caused by peptic ulcer disease, gastric erosions, Mallory-Weiss tears, esophagitis, duodenitis, gastritis, vascular malformations, or malignancy (Alali and Barkun, Gastroenterol Rep (Oxf) 2023, 11: goad011; Barkun et al., Ann Intern Med 2019, 171: 805-22).

21:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

22:

Current guidelines recommend transcatheter arterial embolization (TAE) for patients with upper gastrointestinal (GI) bleeding who cannot tolerate endoscopy or with recurrent gastrointestinal bleeding despite repeated endoscopic therapy (Laine et al., Am J Gastroenterol 2021, 116: 899-917). For lower GI bleeding, possible embolization is recommended if computed tomography angiography (CTA) demonstrates any signs of extravasation of blood (Sengupta et al., Am J Gastroenterol 2023, 118: 208-31).

23:

Any condition which leads to portal hypertension, such as cirrhosis of the liver, can lead to dilatation of the esophageal veins, causing esophageal varices. Patients with chronic cirrhosis and portal hypertension may develop gastric and duodenal varices as well. Patients with varices are at high risk for developing severe bleeding, which can be fatal if left untreated. In cases of variceal hemorrhage, emergency attention must be directed at stopping the blood loss, replacing fluid volume and emergent blood resuscitation, reversal of any coagulation disorders, antibiotic treatment, and antihypertensive treatment to achieve hemodynamic stability. Endoscopy with either banding or sclerotherapy is the most common treatment for esophageal varices. A transjugular intrahepatic portosystemic shunt (TIPS) or surgical shunt placement may be considered for patients not responding to less invasive medical treatment. Adequate management of portal hypertension is essential to prevent refractory bleeding (Henry et al., Clin Gastroenterol Hepatol 2021, 19: 1098-107.e1; Pinchot et al., J Am Coll Radiol 2021, 18: S153-s73; Zanetto and Garcia-Tsao, F1000Res 2019, 8:).

24:

The mainstay of treatment of patients with hemorrhagic shock is blood product transfusion (e.g., packed red blood cells). Despite the general principle of minimizing the use of intravenous fluids in patients with known hemorrhage, crystalloids are often administered until blood products are available, especially in patients who are hypotensive (e.g., mean arterial pressure less than 65). Vital sign changes may underestimate blood loss in patients. Patients with a 30%(0.30) loss of total blood volume may have little to no change in systolic blood pressure (SBP) and only demonstrate modest tachycardia (i.e., heart rate greater than 100). By the time an SBP has dropped to less than 100 mmHg, there may be as much as a 40%(0.40) loss of total blood volume.

25:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).
Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zengin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

26:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).
The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The

terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.
Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

27:

Current guidelines recommend nonemergent versus urgent (within 24 hours) inpatient colonoscopy for patients presenting with acute lower gastrointestinal bleeding (Sengupta et al., Am J Gastroenterol 2023, 118: 208-31).

28:

Upper endoscopy should be considered when a patient has findings of hematemesis or coffee ground emesis. Patients with a history of peptic ulcer disease, liver disease with portal hypertension, and anticoagulant usage or antiplatelet therapy are at greater risk for a brisk upper gastrointestinal bleed. An upper endoscopy is useful for diagnosis and treatment (Laine et al., Am J Gastroenterol 2021, 116: 899-917).

29:

Proton pump inhibitors (PPI) reduce rates of rebleeding and mortality in patients with a bleeding ulcer. Current guidelines recommend continuous or intermittent high-dose PPI, defined as at least 80 mg per day for 3 days, following endoscopic hemostasis in patients with a bleeding ulcer (Laine et al., Am J Gastroenterol 2021, 116: 899-917).

30:

Balloon tamponade is reserved as a bridge to a more definitive surgical or interventional radiologic procedure that will more adequately address the portal hypertension, such as transjugular intrahepatic portosystemic shunt (TIPS) or portosystemic shunt surgery. To prevent the aspiration of blood, intubation may be required for airway protection when undergoing endoscopy or when balloon tamponade is being placed (Bridwell et al., J Emerg Med 2022, 62: 545-58; Henry et al., Clin Gastroenterol Hepatol 2021, 19: 1098-107.e1).

31:

Endoscopy with either endoscopic variceal ligation (EVL) or sclerotherapy is the most common treatment for esophageal varices. Studies have shown a benefit of EVL over sclerotherapy in the initial control of variceal bleeding. If endoscopic or pharmacologic therapy fails to control the bleeding, shunt surgery or transjugular intrahepatic portosystemic shunt (TIPS) has proven efficacy as salvage therapy. To prevent the aspiration of blood, intubation may be required for airway protection when undergoing endoscopy or when balloon tamponade is being performed (Henry et al., Clin Gastroenterol Hepatol 2021, 19: 1098-107.e1; Nett and Binmoeller, Gastrointest Endosc Clin N Am 2019, 29: 321-37; Zanetto and Garcia-Tsao, F1000Res 2019, 8:).

32:

Pharmacological treatment for variceal bleeding: (Cecilio M. Azar and Ala I. Sharara, Conn's Current Therapy 2020. 2020; Zou et al., Ann Transl Med 2019, 7: 717)

33:

Transjugular intrahepatic portosystemic shunt (TIPS) is an interventional radiographic technique in which an intrahepatic metal stent is placed between the intrahepatic portion of the portal and hepatic veins, creating a non-selective portocaval shunt. The TIPS procedure is typically performed in patients who are refractory to medical and endoscopic management (Boike et al., Clin Gastroenterol Hepatol 2022, 20: 1636-62.e36; Pinchot et al., J Am Coll Radiol 2021, 18: S153-s73).

34:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane

Database Syst Rev 2013, 2: CD003408).

35:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.
For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.
NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.
(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

36:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

37:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

38:

Relevant complications or comorbidities are conditions that actively require management during an inpatient stay.

39:

Angiodysplasia is associated with a higher risk of rebleeding in patients with lower gastrointestinal bleeding. These patients may require continued hospitalization for monitoring (Jackson and Strong, Gastrointest Endosc Clin N Am 2017, 27: 51-62; Nishimura et al., Int J Colorectal Dis 2016, 31: 1869-73).

40:

While literature varies in the definition of moderate and severe bleeding, InterQual® external peer reviewers agree that a drop in hemoglobin of 2.0 g/dL(20 g/L) or more in the last 24 hours is appropriate for acute management (Abraham et al., Am J Gastroenterol 2022, 117: 542-58; Dahiya et al., Cleve Clin J Med 2022, 89: 630-3; Gimbel et al., Neth Heart J 2018, 26: 341-51; Valgimigli et al., Eur Heart J 2018, 39: 213-60).

41:

There is a risk for rebleeding following initial treatment of upper gastrointestinal bleeding (Laine et al., Am J Gastroenterol 2021, 116: 899-917; Barkun et al., Ann Intern Med 2019, 171: 805-22).

42:

The American College of Gastroenterology and European Society of Gastrointestinal Endoscopy (ESGE) guidelines recommend repeat endoscopy in patients with recurrent bleeding after endoscopic therapy (Gralnek et al., Endoscopy 2021, 53: 300-32; Laine et al., Am J Gastroenterol 2021, 116: 899-917). Additionally, the ESGE guidelines recommend that routine repeat endoscopy (e.g., scheduled repeat endoscopy typically within 24 hours after initial endoscopy) should be reserved for patients at high risk of recurrent bleeding. Patients who are at high risk of recurrent bleeding include (Gralnek et al., Endoscopy 2021, 53: 300-32):

- Active bleeding lesion at initial endoscopy
- Inability to locate source of bleeding during initial endoscopy
- Incomplete initial endoscopic examination
- Poor initial endoscopic visualization
- Suboptimal endoscopic hemostasis

43:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

44:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO_2) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO_2 with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO_2 retention, where a precise inspired FiO_2 is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO_2. The following are estimates of O_2 delivered at the associated flow rates:
LOW-FLOW SYSTEMS
Nasal cannula
Room air 21%(*0.21*)
1L 24%(*0.24*) **4L** 36%(*0.36*)
2L 28%(*0.28*) **5L** 40%(*0.40*)
3L 32%(*0.32*) **6L** 44%(*0.44*)
Simple oxygen masks
- 35-50%(*0.35-0.50*) FiO_2 (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO_2 in the mask

Venturi masks
- 24-50%(*0.24-0.50*) FiO_2 (4-12 L/min)

Partial rebreathing masks
- 40-70%(*0.40-0.70*) FiO_2 (6-10 L/min)

Non-rebreathing masks
- 60-80%(*0.60-0.80*) FiO_2 (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS
Air-entrainment masks/nebulizers
- 24-40%(0.24-0.40) FiO_2

Heated, humidified high-flow concentrating nasal cannula (HFNC)
- 24-100%(0.24-1.0) FiO_2

45:

Hemodynamic stability is evidenced by adequate tissue and organ perfusion. Blood pressure and heart rate are parameters commonly used to assess hemodynamic stability. Clinical findings associated with hemodynamic stability include normal peripheral pulses, warm extremities, adequate urine output, and neurological stability.

46:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

47:

These findings may be associated with a risk of re-bleeding.

48:

The Child-Pugh classification helps to determine the severity of liver disease. Grades are assigned based on the total points scored: 5-6 points (grade A), 7-9 points (grade B), and 10-15 points (grade C). The points scored are based on the following categories (Daniels et al., In: Henry's Clinical Diagnosis and Management by Laboratory Methods, 24 edn. 2022:314-30):

- Total bilirubin
  - < 2 mg/dL: 1 point
  - 2-3 mg/dL: 2 points
  - > 3 mg/dL: 3 points

- Serum albumin
  - > 3.5 g/dL: 1 point
  - 2.8-3.5 g/dL: 2 points
  - < 2.8 g/dL: 3 points

- Prothrombin time (seconds prolonged) or INR
  - < 4 seconds or INR < 1.7: 1 point
  - 4-6 seconds or INR 1.7-2.3: 2 points
  - > 6 seconds or INR 2.3: 3 points

- Ascites
  - None: 1 point
  - Mild: 2 points
  - Moderate or severe: 3 points

- Hepatic encephalopathy
  - None: 1 point
  - Grade I-II: 2 points
  - Grade III-IV: 3 points

Clinical stages of hepatic encephalopathy include (Dellatore et al., Clin Liver Dis 2020, 24: 189-96; Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47):

- Grade I - Mild confusion, irritability, disordered sleep, decreased attention span
- Grade II - Slurred speech, drowsiness, lethargy, intermittent disorientation, asterixis
- Grade III - Gross disorientation, sleeping but arousable, hyperactive reflexes
- Grade IV - Coma, stuporous, unconscious

49:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

50:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

51:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

52:

A referral should be made to a comprehensive smoking cessation program that includes behavior changing techniques, patient education, and both pharmacological and non-pharmacological interventions (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024).

53:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

54:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

55:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

56:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours

- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

57:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

58:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

59:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.
Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.
Evaluation and Treatment
Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

60:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

61:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

62:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

63:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

64:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

65:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.
The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some

physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

66:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

67:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

68:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

69:

Notable causes for prolonged mechanical ventilation include:
- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):
- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

70:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.